

# Project Proposal

## “Windsor” Integrated Management System — Proposal and Software Requirements Specification

### General Information

**Table 1.** General information

| Field               | Detail                                               |
|---------------------|------------------------------------------------------|
| Project name        | Windsor — Integrated Management System               |
| Development company | General México Software Co.                          |
| Client              | Events-hall chain in the Monterrey metropolitan area |
| Type                | Custom software development (web application)        |
| Budget              | \$1,200,000 MXN                                      |
| Timeline            | 6 months to delivery date                            |
| Document version    | v0                                                   |

### Justification and Objective

The client operates an established chain of event venues (weddings, quinceañeras, corporate events) with multiple spaces and catering service. It currently lacks a unified system to run its operations. The project’s objective is to design, build and deliver a custom integrated management system that centralizes the business —from event booking and supplier services to payments and QR access control— on a single web platform.

Windsor is a web application for booking and managing event halls. It connects clients, service suppliers, administrators, and security staff on a single platform to reserve venues, hire complementary services, manage guest lists, and control access at events through QR codes.

### Project Scope

The project covers seven modules:

1. Authentication and Access Control — registration, login, roles, and password recovery
2. Event Halls — hall catalog, availability, bookings, and maintenance blocking
3. Client Management — client profile, event history, and event management panel
4. Suppliers and Services — supplier service catalog, schedule, and ratings
5. Purchases and Payments — cost calculation, purchase orders, payment status, and receipts
6. Guests and QR Access Control — guest lists, QR generation, and validation at the door
7. Email Notifications — confirmations, QR invitations, schedule changes, and alerts

Each module is described in detail, with its requirements, in the Software Requirements Specification.

## Third-Party Software

Employee management and electronic invoicing (CFDI) are not developed in this project. Invoicing is handled with Facturama, a third-party CFDI service contracted on its annual plan, so the team focuses on the seven modules above. The purchase receipts in RF-COMP-004 are purchase summaries; official invoices are issued through Facturama.

**Table 2.** Third-party usage costs (MXN)

| Service   | Use in the project                              | Plan        | Annual cost |
|-----------|-------------------------------------------------|-------------|-------------|
| Facturama | Electronic invoicing (CFDI) for event purchases | Annual plan | \$1,650 MXN |

## Project Team and Roles

**Table 3.** Project team and roles

| Member                         | Role and skills                                                                                                        | Amount (MXN)         |
|--------------------------------|------------------------------------------------------------------------------------------------------------------------|----------------------|
| <b>General Director</b>        | Owner / founder; developer and client-relationship lead                                                                | N/A (owner)          |
| <b>Project Manager</b>         | Project management and UI/UX design (Figma)                                                                            | \$24,000 MXN         |
| <b>Senior Developer (lead)</b> | Fullstack technical lead; builds the pipeline and coordinates support. Angular, NestJS, TypeScript, PostgreSQL, Docker | \$23,000 MXN         |
| <b>Mid-level Developer</b>     | Fullstack module development (on-site). NestJS, Angular, PostgreSQL                                                    | \$16,000 MXN         |
| <b>Junior Developer</b>        | Module development and support (remote). Angular, TypeScript, Node                                                     | \$14,000 MXN         |
| <b>Monthly team cost</b>       | Sum of fixed monthly salaries (excludes the General Director)                                                          | <b>\$77,000 MXN</b>  |
| <b>Annual team cost</b>        | Monthly team cost × 12                                                                                                 | <b>\$924,000 MXN</b> |
| <b>Total project team cost</b> | Monthly team cost × 6 months                                                                                           | <b>\$462,000 MXN</b> |
| <b>25% profit</b>              | 25% of the total project team cost                                                                                     | <b>\$115,500 MXN</b> |
| <b>Total with 25% profit</b>   | Total project team cost + 25% profit                                                                                   | <b>\$577,500 MXN</b> |

## Proposed Technologies

- **Backend:** TypeScript / NestJS
- **Frontend:** Angular
- **Database:** PostgreSQL
- **Infrastructure:** Docker
- **Internal automation:** a CLI tool (Python) to scaffold backend and frontend and seed the database, reducing repetitive boilerplate.

## Deliverables

- Functional web system with the 7 in-scope modules.
- Modeled and seeded database.
- Development and deployment pipeline (containers).
- Basic technical documentation of the solution.

## Agile Approach and Schedule

The project runs on Scrum: 2-week sprints with incremental delivery. Each sprint closes with a Sprint Review with the client to validate and re-prioritize the Product Backlog; per-sprint ceremonies: planning, daily, review and retrospective. Progress is estimated in user stories and tracked by team velocity.

**Table 4.** Sprint schedule and milestones

| Sprint (2 wks each)            | Sprint goal (increment)                                           | Milestone     |
|--------------------------------|-------------------------------------------------------------------|---------------|
| <b>Sprint 0 (wk 1–2)</b>       | Base architecture, CI/CD pipeline, data model and initial backlog | —             |
| <b>Sprints 1–2 (wk 3–6)</b>    | Authentication and Access Control; Event Halls                    | —             |
| <b>Sprints 3–4 (wk 7–10)</b>   | Client Management; Suppliers and Services                         | —             |
| <b>Sprints 5–6 (wk 11–14)</b>  | Purchases and Payments; Guests and QR Access Control              | MVP release   |
| <b>Sprints 7–8 (wk 15–18)</b>  | Email Notifications and cross-module integration                  | —             |
| <b>Sprints 9–12 (wk 19–26)</b> | Hardening, testing, UAT, training and delivery                    | Final release |

**Definition of Done:** reviewed code (PR), unit and integration tests, and the increment deployed to a QA environment.

**Scrum roles:** Product Owner (prioritizes the backlog and validates with the client), Scrum Master (facilitates and removes impediments) and the Development Team.

## Identified Risks

**Table 5.** Identified risks

| Risk                          | Description                                                                           | Impact   |
|-------------------------------|---------------------------------------------------------------------------------------|----------|
| <b>Key-person dependency</b>  | The project rests on a dual role (coordinator + lead developer) with minimal support. | High     |
| <b>Resource reassignment</b>  | Staff can be moved to other group work without change control.                        | High     |
| <b>PM authority</b>           | The PM tracks status but does not decide on resources; the owner does.                | Med-high |
| <b>Unprotected scope</b>      | No formal change-control process over scope.                                          | High     |
| <b>Client concentration</b>   | A delay from another client can drain this project's resources.                       | Med-high |
| <b>Third-party dependency</b> | Invoicing depends on Facturama, whose prices or availability can change.              | Medium   |

## Assumptions and Notes

- **Link between companies (hypothesis):** it is assumed that General México Software Co. and General Software Company (which has its own internal ERP) are connected by a family relationship (father and son) and share office, infrastructure and staff. This is not confirmed; it is used as an explanatory hypothesis.
- **Team salaries:** salaries are fixed monthly amounts for illustration.
- **Schedule (Scrum):** the 6-month sprint plan is a proposed reconstruction, not necessarily the original formal plan.
- **Actual project status:** in real execution, the project was frozen midway and the lead technical resource was reassigned to another system in the group. This proposal reflects the project as it should have been planned, not how it ended.

# Software Requirements Specification

## Module Overview

### Authentication and Access Control

This module manages user registration and login for clients and suppliers, using JWT tokens to keep sessions secure. It also restricts what each user can see and do based on their role (Administrator, Client, Supplier, or Security Guard), and lets users recover their password through a temporary token sent by email.

### Event Halls

This module lets the Administrator register event halls with their name, location, maximum capacity, and hourly cost, upload photos of each hall, and check availability by date and time range. It also keeps a booking schedule that prevents overlapping or duplicate reservations, and allows the Administrator to block dates or time slots for maintenance or private events.

### Client Management

This module gives clients a profile where they can view and update their personal data, contact information, and notification preferences. Clients can also see a detailed history of their past, active, and cancelled events with invoices or summaries, and manage their guest list and hired supplier services from a dedicated panel for each active event.

### Suppliers and Services

This module allows suppliers to register and update the services they offer (catering, entertainment, music, decoration, etc.) along with descriptions and prices, and to attach one or more of these services to a hall reservation. Suppliers can view a calendar with the dates, times, and halls where they must deliver their services, and clients can rate and review a supplier once an event has concluded.

### Purchases and Payments

This module calculates the total cost of an event by adding the hall's hourly cost to the accumulated cost of the selected supplier services. It generates and stores a purchase order linked to the client and the event, allows the Administrator or the payment gateway to update the payment status (Pending, Completed, Refunded), and produces a detailed digital receipt that the client can view or download.

### Guests and QR Access Control

This module lets clients add guests one by one or through a bulk upload with contact details, and automatically generates a unique QR code for each guest and event. Security guards use the web application to scan and validate each QR code at the door; once a code is successfully scanned it is marked as "Used" so it cannot be reused, and clients can resend a guest's QR code individually if it gets lost.

## Email Notifications

This module automatically emails clients to confirm their hall booking and hired services, sends each registered guest their QR-coded invitation, and notifies clients, suppliers, and guests by email whenever an event's date changes or the event is cancelled. It also alerts the Administrator when a new supplier requests registration or when an external integration fails.

## Functional Requirements

**Table 6.** Functional requirements

| ID          | Requirement                   | Description                                                                                                                                      |
|-------------|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| RF-AUTH-001 | User registration             | The system must allow clients and suppliers to register by entering their basic information and access credentials.                              |
| RF-AUTH-002 | JWT authentication            | The system must validate user credentials and generate a JWT token to manage secure sessions.                                                    |
| RF-AUTH-003 | Role-based access control     | The system must restrict views and actions according to the assigned role (Administrator, Client, Supplier, Security Guard).                     |
| RF-AUTH-004 | Password recovery             | The system must allow users to recover access through a temporary token sent to their registered email.                                          |
| RF-SAL-001  | Event hall registration       | The system must allow the Administrator to register halls, defining name, location, maximum capacity, and hourly cost.                           |
| RF-SAL-002  | Hourly availability lookup    | The system must allow checking a hall's availability on any date of the year, filtering by specific time ranges.                                 |
| RF-SAL-003  | Booking schedule              | The system must record bookings by time range, preventing overlap or duplicate schedules for the same hall.                                      |
| RF-SAL-004  | Hall image upload             | The system must allow uploading and storing photographs of the halls.                                                                            |
| RF-SAL-005  | Date blocking for maintenance | The system must allow the Administrator to disable halls or specific time ranges on given days for maintenance or private events.                |
| RF-CLI-001  | Client profile management     | The system must allow the Client to view and update their personal data, contact information, and notification methods.                          |
| RF-CLI-002  | Event history                 | The system must show the Client a detailed list of their past, active, and cancelled events with their respective invoices or summaries.         |
| RF-CLI-003  | Event management panel        | The system must provide the Client with a panel to manage the guest list and the hired supplier services for an active event.                    |
| RF-PROV-001 | Supplier service registration | The system must allow Suppliers to register and update their services (catering, entertainment, music, decoration) with descriptions and prices. |
| RF-PROV-002 | Linking services to events    | The system must allow attaching one or more supplier services to an event hall booking.                                                          |
| RF-PROV-003 | Supplier schedule lookup      | The system must offer Suppliers a calendar to view the dates, times, and halls where they must deliver their services.                           |

| ID                 | Requirement                        | Description                                                                                                                              |
|--------------------|------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| <b>RF-PROV-004</b> | Supplier rating                    | The system must allow the Client to rate and leave a review of the services provided by the Supplier once the event has concluded.       |
| <b>RF-COMP-001</b> | Total purchase cost calculation    | The system must calculate the total amount by adding the hourly hall cost and the accumulated cost of the selected supplier services.    |
| <b>RF-COMP-002</b> | Purchase order registration        | The system must generate and store the purchase order in pending or paid status, associating it with the client and the event.           |
| <b>RF-COMP-003</b> | Transaction status change          | The system must allow the Administrator or the payment gateway to update the payment status (Pending, Completed, Refunded).              |
| <b>RF-COMP-004</b> | Purchase receipt generation        | The system must generate a detailed summary of the purchase in digital format for the client to view or download.                        |
| <b>RF-INV-001</b>  | Guest list upload                  | The system must allow the Client to enter guests individually or perform a bulk upload with contact details.                             |
| <b>RF-INV-002</b>  | Unique QR code generation          | The system must generate a unique hash and represent it graphically as an exclusive QR code for each guest and registered event.         |
| <b>RF-INV-003</b>  | QR scanning and validation         | The system must allow the Security Guard to validate the authenticity of the QR code presented by the guest through the web application. |
| <b>RF-INV-004</b>  | Real-time access control           | The system must mark the QR code as “Used” after the first successful scan to prevent the same entry from being reused.                  |
| <b>RF-INV-005</b>  | QR code resend                     | The system must allow the Client to individually resend the QR-coded ticket if a guest loses it.                                         |
| <b>RF-MAIL-001</b> | Purchase confirmation notification | The system must send an automatic email to the Client confirming the hall booking and the hired services.                                |
| <b>RF-MAIL-002</b> | Bulk QR access delivery            | The system must automatically send invitations with the packaged QR code to the email address of each registered guest.                  |
| <b>RF-MAIL-003</b> | Schedule change notification       | The system must notify Clients, Suppliers, and Guests by email of any change in date or cancellation of the event.                       |
| <b>RF-MAIL-004</b> | Administration alerts              | The system must notify the Administrator when a new Supplier requests registration or when a failure occurs in external integrations.    |

## Non-Functional Requirements

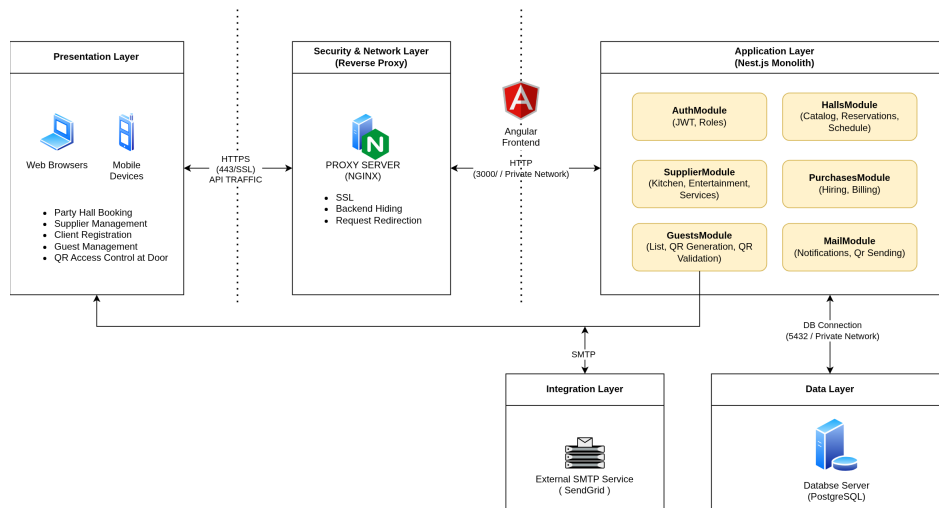
**Table 7.** Non-functional requirements

| ID           | Requirement               | Description                                                                                                                             |
|--------------|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| RNF-SEG-001  | Reverse proxy concealment | The NestJS monolithic backend must operate behind a Proxy server (Nginx) without publicly exposing its private IP or ports.             |
| RNF-SEG-002  | Database isolation        | The database must be hosted on a physically independent server with no public access, accepting connections only from the backend's IP. |
| RNF-PER-001  | API response time         | Requests processed by the NestJS routes must respond with an average time under 300 ms under normal traffic conditions.                 |
| RNF-USAB-001 | Responsive interface      | The Angular web application must adapt its layout to desktop browsers and mobile devices to facilitate QR reading at the door.          |
| RNF-DISP-001 | Service availability      | The system must maintain 99.5% operational availability throughout the year to guarantee real-time queries and validations.             |
| RNF-COMP-001 | Scanning compatibility    | The QR validation module must be compatible with the cameras of mobile devices running modern browsers (Chrome, Safari).                |

## Architecture

**Figure 1** shows the architecture of the party hall management system, and **Table 8** summarizes its layers.

**Windsor:**  
**Party Hall Management System Architecture**



**Figure 1.** Windsor: party hall management system architecture

**Table 8.** Architecture layers

| Layer                         | Components                                                                                                                  | Responsibilities and connections                                                                                                                                                                              |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Presentation</b>           | Web browsers, mobile devices                                                                                                | Party hall booking, supplier management, client registration, guest management, and QR access control at the door. Sends API traffic over HTTPS (443/SSL).                                                    |
| <b>Security &amp; Network</b> | Nginx proxy server (reverse proxy)                                                                                          | SSL, backend hiding, and request redirection. Forwards traffic over HTTP (3000 / private network).                                                                                                            |
| <b>Application</b>            | NestJS monolith: AuthModule, HallsModule, SupplierModule, PurchasesModule, GuestsModule, MailModule                         | JWT and roles; catalog, reservations, and schedule; supplier services (kitchen, entertainment, and services); hiring and billing; guest list, QR generation, and QR validation; notifications and QR sending. |
| <b>Integration</b>            | External SMTP service (SendGrid); external payment gateway (vendor to be defined); electronic invoicing service (Facturama) | Email delivery over SMTP; payment status updates for purchase orders (RF-COMP-003); official invoices for purchases.                                                                                          |
| <b>Data</b>                   | Database server (PostgreSQL); file storage                                                                                  | Persistence over a DB connection (5432 / private network). Hall photographs (RF-SAL-004) are stored by the application in file storage, with their references in PostgreSQL.                                  |

**Deployment:** the services run in Docker containers, and the CI/CD pipeline deploys each increment to a QA environment before delivery, as required by the Definition of Done in Agile Approach and Schedule.